

Chapter: Merge Strings Alternately

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Visualizing the Concept

Example Walkthrough

WORD1 (AR FINISHES)

a r

WORD2 (AADYA)

a a d y a

MERGED OUTPUT

a a r a d y a

"aa is only proccessed, dya remains"

Logic & Algorithm Notes

Main Loop Condition

```
while(idx1<n1 && idx2<n2)
```

Rule: "if any string finishes break off loop". We alternate characters until one of the strings runs out.

Alternating Turns

Using the counter check: `if(cnt%2==1)`

Rule: "if odd word1 se lenge". If it is an odd turn, we append a character from `word1` and increment our indices. Otherwise, we take from `word2`.

Handling Leftovers

"for leftovers

eg

ar

aadya

ar finishes

aa is only proccessed

dya remains"

C++ Source Code

```
class Solution {
public:
    string mergeAlternately(string word1, string word2) {

        int cnt = 1;
        int idx1 = 0;
        int idx2 = 0;
        int n1 = word1.size();
        int n2 = word2.size();
        string ans = "";

        while(idx1 < n1 && idx2 < n2){ //if any string finishes break off loop
            if(cnt % 2 == 1){
                //if odd word1 se lenge
                ans += word1[idx1];
                cnt++;
                idx1++;
            }

            else{
                ans += word2[idx2];
                cnt++;
                idx2++;
            }
        }

        //for leftovers
        //eg
        //ar
        //aadya

        //ar finishes
        //aa is only proccessed
        //dya remains

        while(idx1 < n1){
            ans += word1[idx1];
            idx1++;
        }

        while(idx2 < n2){
```